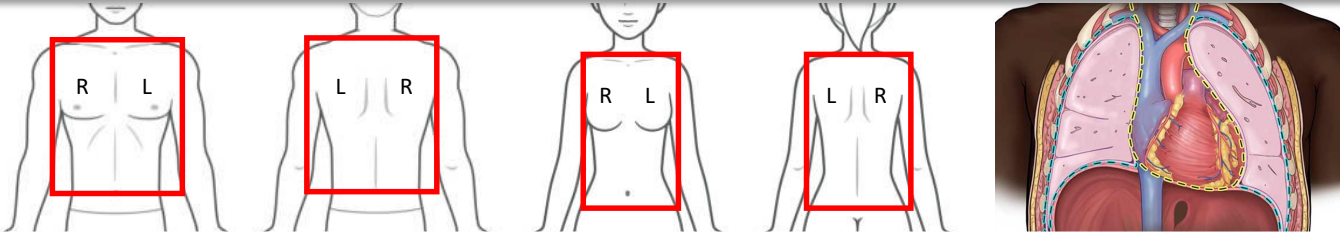


CHEST INJURIES

Immediate life-threatening injuries should be addressed in the 'BREATHING' aspect of the primary survey. BLATOMFC is the mnemonic used for all life-threatening traumatic chest injuries.

MANAGE HAEMORRHAGE, AIRWAY & ASSESS CONSCIOUS LEVELS

THE CRITICAL BOX - CONTAINS MAJOR ORGANS & BLOOD VESSELS



IS THERE ANY PENETRATING TRAUMA TO THE CRITICAL BOX?

YES

NO

VITALS (RESPONSE, BREATHING, PULSES) PRESENT?

YES

NO

CHECK FOR ENTRY AND EXIT WOUNDS

PLACE CHEST SEALS THEN FULL ASSESSMENT WITH MONITORING

TRAUMATIC ARREST (HOTT APPROACH)

RESUSCITATIVE THORACOTOMY (pg 25) IF: WITHIN 15 MINUTES OF LOSS OF VITAL SIGNS

MOVE TO ASSESSMENT
CONSIDER SPINAL PROTECTION IF ENVIRONMENT ALLOWS & MECHANISM FITS WITH: REDUCED CONSCIOUSNESS and/or NEUROLOGICAL DEFICIT (MOTOR OR SENSORY)

BREATHING/RESPIRATORY ASSESSMENT (INITIAL LOOK)

RESPIRATORY FAILURE or RESPIRATORY ARREST

AIRWAY MANAGEMENT AND SUPPORT VENTILATIONS

FULLY ASSESS THORAX, BACK AND ARMPITS FOR INJURIES AND RESUSCITATE (OXYGENATE AND VOLUME)

PENETRATING or BLAST (DEBRIS) TRAUMA

EARLY ROLL TO CHECK ALL THORACIC REGION INCLUDING BACK AND ARMPITS

TREAT OPEN CHEST WOUNDS PREPARE FOR MEDEVAC (STRETCHER AND HPMK)

BLUNT TRAUMA (RTC, FALL, EJECTED)

CONSIDER C-SPINE PROTECTION

FULL BREATHING ASSESSMENT

DELAY BACK CHECK WHEN ABLE TO LOG ROLL PATIENT AND PREPARE FOR MEDEVAC

REGULAR REASSESSMENT IS REQUIRED IN PATIENTS WITH SIGNIFICANT EXTERNAL CHEST INJURIES AS THERE IS RISK OF INTERNAL INJURY THAT CAN CLINICALLY DEVELOP OVER TIME. DON'T DELAY MEDEVAC!

BREATHING/RESPIRATORY MANAGEMENT

BREATHING RATE <10	VENTILATE PATIENT IF IN RESPIRATORY FAILURE/FATIGUE (POOR EFFORT or RR <10) MAINTAIN NORMAL END-TIDAL CO ₂ (4-5.7kPa or 35-45mmHg) IF USING ADVANCED AIRWAY WITH CIRCUIT (FOLLOW HEAD INJURY GUIDANCE IN THE SETTING OF TBI)
BREATHING RATE >24	HIGH FLOW OXYGEN IF CRITICALLY INJURED OR UNWELL (DO NOT WAIT FOR O ₂ SATS) PROLONGED CARE - TITRATE OXYGEN TO MAINTAIN OXYGEN SATURATIONS >95% REGULAR MONITORING FOR DETERIORATION (TENSION, HYPOXIA, HYPOVOLAEMIA)
CIRCULATION	CONSIDER HAEMOTHORAX, TENSION PNEUMOTHORAX OR CARDIAC TAMPONADE IN SHOCKED PATIENTS. RESUSCITATE WITH BLOOD. FLAT NECK VEINS = VOLUME PROBLEM.

LIFE-THREATENING CHEST INJURIES - BLATOMFC

BLAST LUNG	PRESENTATION: DYSPNOEA, CHEST PAIN, HAEMOPTYSIS POST BLAST IN CONFINED AREA TREATMENT: OXYGENATE, ANALGESIA AND ASSESS FOR WORSENING PATHOLOGY
AIRWAY OBSTRUCTED	PRESENTATION: FLUIDS, SOLIDS, STRUCTURAL/MAXILLOFACIAL DAMAGE OR LOW GCS TREATMENT: AIRWAY LADDER APPROACH (FOLLOW ADVANCED AIRWAY GUIDANCE)
TENSION PNEUMO-THORAX	PRESENTATION: PENETRATING OR BLAST INJURIES, OR PPV PATIENTS = RESPIRATORY AND CIRCULATORY COLLAPSE DUE TO INCREASING INTRATHORACIC PRESSURES. TREATMENT: OXYGENATE AND DECOMPRESS WITH NEEDLE THORACOSTOMY. THIS SHOULD BE UPGRADED TO TUBE THORACOSTOMY IN PFC OR PROLONGED MEDEVAC.
OPEN PNEUMO-THORAX	PRESENTATION: PENETRATING OR BLAST INJURIES = RESPIRATORY IMPAIRMENT WITH OPEN WOUND TO THORAX. EARLY BACK ASSESSMENT IS REQUIRED. TREATMENT: CHEST SEAL PER HEMISPHERE APPLIED UPPERMOST AND OXYGENATE. CONSIDER OTHER SIGNIFICANT INTERNAL INJURIES (TENSION OR HAEMOTHORAX)
MASSIVE HAEMO-THORAX	PRESENTATION: CONSIDER IN SHOCKED PATIENT WITH EXTENSIVE OR PENETRATING CHEST INJURIES. TREATMENT: SEAL OPEN WOUNDS, OXYGENATE AND RESUSCITATE WITH BLOOD TO MAINTAIN NORMAL MENTATION WHILST TRANSFERRING TO SURGICAL CAPABILITY. ONLY CONSIDER TUBE THORACOSTOMY IF THERE IS SIGNIFICANT RESPIRATORY IMPAIRMENT.
FLAIL CHEST (FRACTURED RIBS)	PRESENTATION: SEVERE PAIN ON BREATHING WITH PARADOXICAL CHEST MOVEMENT FROM A SEGMENT OF MULTIPLE ADJACENT FRACTURED RIBS. TREATMENT: OXYGENATE TO MAINTAIN O₂ SATS >95% AND STRONG ANALGESIA TO ALLOW PATIENT TO EFFECTIVELY BREATHE SPONTANEOUSLY. USE ANALGESIA LADDER.
CARDIAC TAMPONADE	PRESENTATION: MAINLY PENETRATING TRAUMA TO ANTERIOR/LATERAL CHEST WITH SHOCK PHYSIOLOGY, INCREASED JVD AND MUFFLED HEART SOUNDS WITH PROGRESSIVE DETERIORATION DESPITE BLOOD RESUSCITATION. TREATMENT: TREAT AS HYPOVOLAEMIA IN SHOCKED PATIENTS, URGENT MEDEVAC TO SURGICAL CAPABILITY. IF IN TCA, CONSIDER RESUSCITATIVE THORACOTOMY.

POINTS TO NOTE

1. IF SEVERE DYSPNOEA OCCURS, ALLOW THE PATIENT TO BE IN A POSITION THAT SUPPORTS THEIR OWN BREATHING, WHICH IS USUALLY SAT UP. IF THERE IS CONCERN FOR C-SPINE, MAINTAIN THE HEAD AND NECK IN A NEUTRAL POSITION AND PREPARE PACKAGING AROUND THE PATIENT.
2. DO NOT DELAY OXYGENATION TO WAIT FOR AN OXYGEN SATURATIONS READING IN TRAUMATIC CHEST INJURIES. TITRATION CAN OCCUR IN THE PFC PHASE.
3. IF PERFORMING A LOG ROLL TO CHECK THE BACK, IT IS PRACTICAL TO GET THE STRETCHER, BLIZZARD BLANKET AND PELVIC BINDER TO THE SIDE OF THE PATIENT SO IT CAN BE PLACED UNDER THE IMMOBILISED PATIENT TO AVOID FURTHER UNNECESSARY ROLLS.
4. A PATIENT WITH A TENSION PNEUMOTHORAX WILL BE CRITICALLY UNWELL WITH BREATHING SIGNS AND SHOCK. THEY ARE LIKELY TO REQUIRE CRITICAL CARE MANAGEMENT WITH THORACOSTOMY SO EARLY EVACUATION IS REQUIRED.
5. TUBE THORACOSTOMY IS LIKELY REQUIRED IN PROLONGED TRANSFER OR HOLDS OF SPONTANEOUSLY BREATHING TENSION PNEUMOTHORAX PTS.
6. RIB FRACTURES REQUIRE STRONG ANALGESIA TO ALLOW FOR APPROPRIATE DEPTH IN BREATHING. OPIOIDS ARE RECOMMENDED FOR MODERATE TO SEVERE PAIN AND WOULD BE BENEFICIAL IN THESE PATIENTS.
7. CONSIDER PULMONARY CONTUSIONS IN PATIENTS WITH BLAST OR BLUNT TRAUMA WHICH MAY RESULT IN DYSPNOEA AND HYPOXIA. OXYGENATION AND MEDEVAC ARE REQUIRED IN THESE PATIENTS.

Guideline written in accordance with SFM Clinical Standard Operating Guideline (CSOG) 003 'Management of Chest Injuries' (2022)

References

- Leech, C., et al. 2019. The Pre-Hospital Management of Life-Threatening Chest Injuries: A Consensus Statement from the Faculty of Pre-Hospital Care, Royal College of Surgeons of Edinburgh, *Trauma*, 19(1), pp. 54-62.
- Ludwig, C. and Koryllos, A. 2017. Management of Chest Trauma, *Journal of Thoracic Disease*, 9(Suppl 3), pp. S172-177.